

Network fundamentals continues :-

IPv4 addresses are limited (4.3 billion total) because of shortage. we split them into 2 categories :-

1) Private IPv4 :- used inside your home or office, these addresses cannot be seen by the Internet. Ex:- 192.168.XX, 10.X.X.X or 172.16.X.X.

Public IPv4 :- Assigned to router by your ISP. These show rest of world finds your network.

Ex:- Cisco an industry giant in world of networking, estimated there would be approximately 50 billion devices connected ^{to internet} on end of 2021. So far we discussed on version of IPv4 Internet protocol.

which use 2^{32} IP Address (4.29 billion) - See why there is a shortage.

* IPv6 is new generation of Internet protocol to help tackle the issue.

→ supports upto 2^{128} IP addresses (340 trillion plus, resolving issues faced on IPv4.

→ More efficient due to new methodologies

IPv6 address: 2a00:22c4:a531:c500:425f:cee6:c36b:f64d

IPv4 address: 86-157.52.21

Public IPv6 address the goal is every device has its own unique public address. These usually starts with 2 or 3 ex:- (2001:0db...)

Private IPv6 (unique local) :- Even we have plenty of addresses, organizations still use private addresses for internally security starts with fd or fc.

MAC Addresses :-

Devices on network will have a physical ^{network} interface, which is a microchip board found on device's motherboard. This network

Interface is assigned a unique address at the factory it was built at, called a MAC (Media Access Control) Address. The MAC address is a twelve character (12) hexadecimal number split into two's and separated by a colon. These colons are considered separators.
Ex:- a4:c3:f0:85:ac:2d. The first 6 characters represent

Vendor who build the network Interface

unique address of the network Interface

a4:c3:f0:85:ac:2d

2. MAC Spoofing (The "Fake ID") :-

An interesting thing with MAC addresses is they can be faked or "spoofed" in process known as spoofing. This spoofing occurs when a network device pretends to identify as another using its MAC address. When this occurs, it can often break poorly implemented security design that assume that devices talking on network are trustworthy.

Ex:- It's like wearing high quality silicone mask of someone's else face.
→ If a firewall only lets 'manager' access a network, a hacker can spoof managers MAC address to trick system into letting them in.
→ The system thinks it's talking to manager, but it's actually talking to an imposter.

Ex 2:- A firewall is configured to allow communication going to and from MAC address to administrator. If device were pretend or "spoof" this MAC address, the firewall would now think that it is receiving communication from Administrator where it isn't.

places such as cafes, coffee shops and hostels alike often use MAC address control when using their "Guest" or "Public" wifi, practical lab in github:-

The Interactive labs simulate a hotel wifi network where they have to pay for the service. you will see that router is not allowing Bob's packets (blue) to TryHackMe website and is placing them in bin, but Alice packets (green) are going through fine because she paid for wifi. Try changing Bob's Mac Address same as Alice's to see what happens - (then it works fine).

Q1) What does IP stand for? Q2) What is each section of IP address called?

A) Internet Protocol

A) Octet

Q3) How many sections (in digits) does an IP address have?

A) 4

Q4) What does term "MAC" stand for?

A) Media Access Control

Q5) Deploying lab what is the flag?

A) THM\$YOU-GOT-ON-TRYHACKME\$